PHYS 143 – Mechanics and Thermal Physics

Week 2: Motion – Supplementary Notes

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These notes support the Week 2 lecture. They contain derivations and background that were moved out of the contact time. Read them after the lecture; they are not needed to follow the lecture problems, except Section 1 if the Week 2 diagnostic showed gaps in calculus.

1 Calculus refresher

Derivatives. For a constant c and a power n:

$$\frac{\mathrm{d}}{\mathrm{d}t}\,c = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\,t^n = nt^{n-1}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\cos\omega t = -\omega\sin\omega t, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\sin\omega t = \omega\cos\omega t.$$

The derivative of a sum is the sum of the derivatives, and a constant factor comes out. Example: $x = 3t^2 - 4t + 2$ gives $\mathrm{d}x/\mathrm{d}t = 6t - 4$.

Integrals. The reverse operation:

$$\int t^n\,\mathrm{d}t = \frac{t^{n+1}}{n+1} + C \quad (n \neq -1), \qquad \int \cos\omega t\,\mathrm{d}t = \frac{\sin\omega t}{\omega} + C, \qquad \int \sin\omega t\,\mathrm{d}t = -\frac{\cos\omega t}{\omega} + C.$$

In physics the constant C is fixed by an initial condition. Example: $a = 6t$ with $v = 0$ at $t = 0$: $v = 3t^2 + C$, and $C = 0$.

2 Derivation of the constant-acceleration equations

Let a_x be constant, with $v_x = v_{0x}$ and $x = x_0$ at $t = 0$.

Velocity and position by integration

$$v_x = v_{0x} + \int_0^t a_x\,\mathrm{d}t' = v_{0x} + a_x t \qquad (1)$$

$$x - x_0 = \int_0^t (v_{0x} + a_x t')\,\mathrm{d}t' = v_{0x}t + \tfrac{1}{2}a_x t^2 \qquad (2)$$

Equation (3), no time. From (1), $t = (v_x - v_{0x})/a_x$. Substitute into (2):

$$x - x_0 = v_{0x}\frac{v_x - v_{0x}}{a_x} + \frac{(v_x - v_{0x})^2}{2a_x} = \frac{v_x^2 - v_{0x}^2}{2a_x} \;\Rightarrow\; v_x^2 = v_{0x}^2 + 2a_x(x - x_0).$$

Equation (4), no acceleration. Because v_x changes linearly with t, the average velocity over the interval is $\frac{1}{2}(v_{0x} + v_x)$, so $x - x_0 = \frac{1}{2}(v_{0x} + v_x)\,t$.

Equation (5), no initial velocity. Substitute $v_{0x} = v_x - a_x t$ into (2): $x - x_0 = (v_x - a_x t)t + \frac{1}{2}a_x t^2 = v_x t - \frac{1}{2}a_x t^2$.

3 Height, time of flight and range of a projectile

Launch from level ground with speed v_0 at angle θ above the horizontal, and $+y$ upward:

$$v_{0x} = v_0\cos\theta, \qquad v_{0y} = v_0\sin\theta.$$

Time to the top. $v_y = 0$ gives $t_{\text{top}} = v_0 \sin\theta / g$.

Maximum height. Equation (3) in the y-direction:

$$y_{\max} = \frac{v_0^2 \sin^2\theta}{2g}.$$

Time of flight. Setting $y = 0$ in $y = v_{0y}t - \frac{1}{2}gt^2$ gives $t = 0$ or $T = 2v_0 \sin\theta / g = 2t_{\text{top}}$.

Range.

$$R = v_{0x}T = \frac{2v_0^2 \sin\theta\cos\theta}{g} = \frac{v_0^2 \sin 2\theta}{g}.$$

The range is greatest at $\theta = 45°$ and has the same value for θ and $90° - \theta$. These formulas hold only for equal launch and landing heights; for other cases solve the quadratic in t directly, as in the lecture.

Check with the lecture example

$v_0 = 30.0\,\text{m/s}$, $\theta = 40.0°$: $R = (900/9.80)\sin 80° = 90.4\,\text{m}$, $y_{\max} = (900\sin^2 40°)/(2\times 9.80) = 19.0\,\text{m}$.

4 The trajectory equation

Eliminating $t = x/(v_0\cos\theta)$ from $y = v_0\sin\theta\, t - \frac{1}{2}gt^2$:

$$y = x\tan\theta - \frac{g}{2v_0^2\cos^2\theta}\,x^2,$$

a parabola. This is the general parametric-to-Cartesian step shown in the lecture.

5 The helix from the old Lecture 2 deck

$\vec{\boldsymbol{r}} = a\cos\omega t\,\hat{\boldsymbol{\imath}} + b\sin\omega t\,\hat{\boldsymbol{\jmath}} + ct^2\,\hat{\boldsymbol{k}}$. Differentiating:

$$\vec{\boldsymbol{v}} = -a\omega\sin\omega t\,\hat{\boldsymbol{\imath}} + b\omega\cos\omega t\,\hat{\boldsymbol{\jmath}} + 2ct\,\hat{\boldsymbol{k}}, \qquad \vec{\boldsymbol{a}} = -a\omega^2\cos\omega t\,\hat{\boldsymbol{\imath}} - b\omega^2\sin\omega t\,\hat{\boldsymbol{\jmath}} + 2c\,\hat{\boldsymbol{k}}.$$

The xy-projection is an ellipse (a circle if $a = b$) and the z-motion has constant acceleration $2c$. The acceleration in the xy-plane is $-\omega^2$ times the xy-position: it points to the axis, as in uniform circular motion.

6 Motion in a circle: a preview of dynamics

In the lecture we only used kinematics: for uniform circular motion $a_{\text{rad}} = v^2/R$ toward the centre. Because $\vec{\boldsymbol{a}} \neq 0$, Newton's second law implies that a *net force* toward the centre is required. That force is not a new kind of force; it is whatever real force (tension, friction, gravity, a normal force) happens to point to the centre. This is developed in the Forces and Circular motion topics.

7 Air resistance, in words

Every example in the lecture neglects air resistance. In reality a drag force opposes the velocity and increases with speed. The x- and y-motions of a real projectile are then no longer independent, the range is shorter, and the descent is steeper than the ascent. A falling object eventually reaches a *terminal speed* where drag balances weight and $a_y \to 0$. None of this is needed for Week 2.